

Multimodal Hate Video Classification on HateMM

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Problem definition

Problem. Hateful content online increasingly travels as video, where meaning is carried jointly by visual context, audio delivery, and the spoken message. Single-modality classifiers struggle on cases where these three signals must be combined to distinguish hate from satire or critical discourse.

Hypothesis. A small transformer trained on top of frozen pretrained per-modality encoders can outperform every unimodal baseline on a real corpus of hateful and non-hateful videos, with gains that are not uniform across target communities.

Dataset

HateMM (Das et al., ICWSM 2023), released under CC BY 4.0 on Zenodo. We use the corpus as-is, without modification or redistribution.

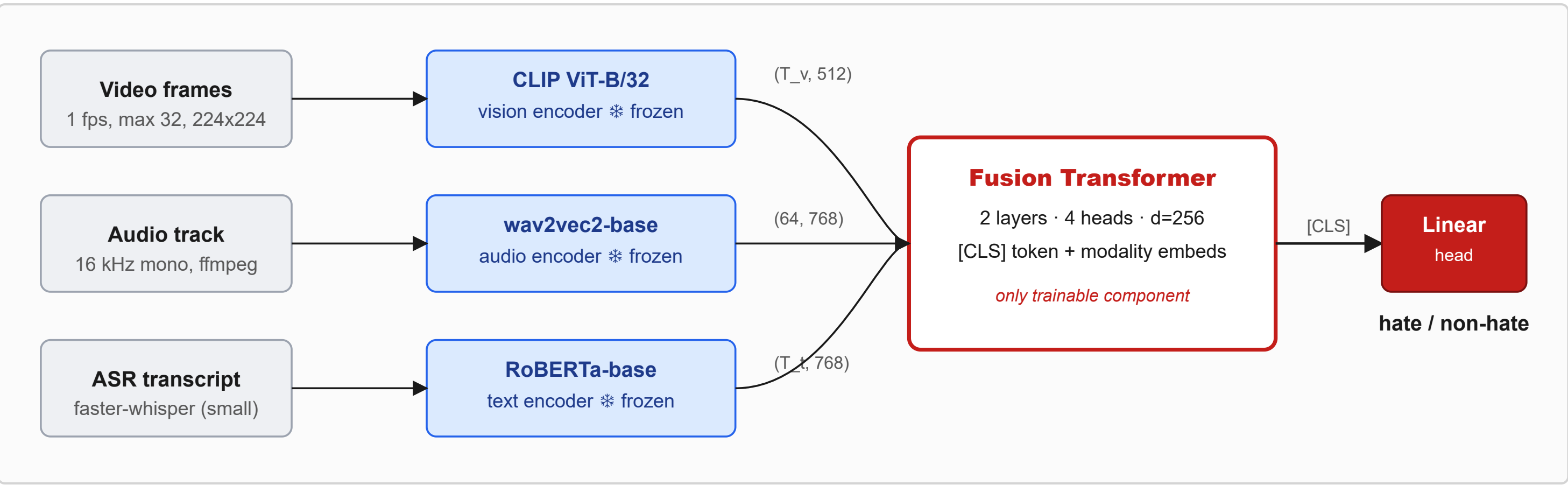
- **1083 BitChute videos:** 431 hate, 652 non-hate.
- Approximately 43.5 hours of content, 6.3 GB on disk.
- Per-video labels: binary hate flag, target community, hateful-segment timestamps.
- Stratified split by label, fixed seed: 80 % train (863), 10 % val (109), 10 % test (109).
- Preprocessing succeeded on 1081 / 1083 videos.

Method

Pipeline in three stages.

1. **Preprocess.** Sample frames at 1 fps capped at 32 per video, center-cropped to 224 x 224; extract 16 kHz mono audio via ffmpeg; transcribe speech with faster-whisper.
2. **Frozen feature extraction.** CLIP ViT-B/32 per frame, wav2vec2-base mean-pooled into 64 audio tokens, RoBERTa-base token embeddings (length 128) on the transcript.
3. **Fusion.** A 2-layer transformer (d = 256, 4 heads, GELU, pre-norm) ingests the concatenated per-modality token sequences plus learnable modality-type embeddings and a [CLS] token; the [CLS] output goes through a linear head for binary classification.

Why frozen encoders. Keeping all three encoders frozen makes the modality ablation a fair comparison: only the small fusion head varies between configurations. Restricting the same architecture to a single modality at the input gives our unimodal baselines.



Related works

- **HateMM [1].** Defines the task; releases the dataset and target-community labels we audit against.
- **CLIP [2].** Provides our frozen vision encoder; we use the image branch only.
- **Multimodal fusion transformers [3].** Motivate the cross-modal-attention design; we adopt a lighter [CLS]-readout variant suited to the dataset scale.

Validation

Table 1. Modality ablation on the test split (n = 109).

Modality	Acc	F1	Prec	Rec	AUROC
V + A + T	0.835	0.824	0.821	0.744	0.873
Text only	0.780	0.763	0.757	0.651	0.842
Video only	0.743	0.726	0.692	0.628	0.815
Audio only	0.697	0.695	0.589	0.767	0.775

Table 2. Per-target-community test accuracy (n ≥ 10).

Community	n	Fusion	Video	Audio	Text
Blacks	46	0.804	0.717	0.630	0.717
Others	39	0.846	0.795	0.718	0.872
Jews	15	0.867	0.600	0.733	0.733

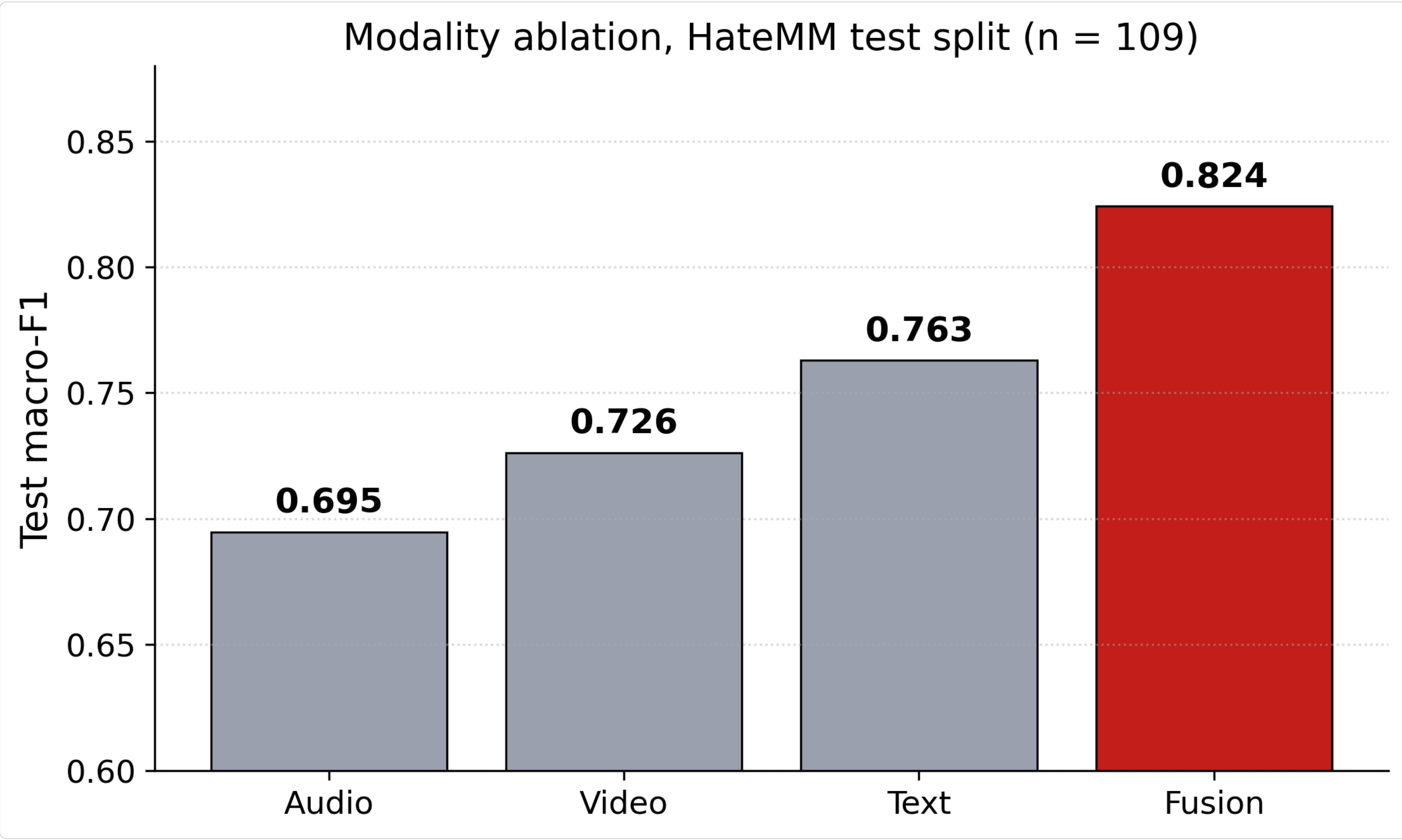


Figure 1. Test macro-F1, four-model modality ablation.

Key observations

Fusion wins on every aggregate metric except recall, where audio-only scores higher; audio behaves as a high-recall, low-precision signal that fusion integrates well.

Text is the strongest unimodal baseline, consistent with the verbal nature of much hate speech.

On *Others*, text alone slightly beats fusion; on *Jews* (smallest reported group), fusion shows the largest gain over any single modality.

Limitations

- **Single seed.** Headline numbers come from one run; multi-seed averages would tighten the headline gap.
- **Small test set (n = 109).** Per-community statistics for groups with n < 10 are excluded as unreliable.
- **ASR quality varies.** Shouting, music, and accented speech degrade the text modality on a subset.
- **Frozen backbones cap absolute performance.** Intentional methodological choice that keeps the modality ablation interpretable.
- **Class imbalance not loss-weighted.** Recall on the hate class is 0.744; the model is conservative on positives.

Conclusion

Fusing three frozen pretrained encoders through a small transformer head outperforms every unimodal baseline on HateMM hate-video classification by a clean margin: +6 points in macro-F1 and +3 points in AUROC over the best unimodal. The per-community audit reveals that the gain is not uniform: text alone suffices for some target groups while fusion is decisive for others, with direct implications for moderation pipelines that today combine modality-specific classifiers via simple voting.

Next steps. Multi-seed averaging of the headline table, a counterfactual modality-masking probe for per-example modality attribution, and class-rebalanced training to lift hate recall.

References

1. M. Das, R. Raj, P. Saha, B. Mathew, M. Gupta, and A. Mukherjee, "HateMM: A Multi-Modal Dataset for Hate Video Classification," in *Proc. AAAI ICWSM*, vol. 17, 2023, pp. 1014-1023.

2. A. Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," in *Proc. ICML*, vol. 139, 2021, pp. 8748-8763.

3. A. Nagrani, S. Yang, A. Arnab, A. Jansen, C. Schmid, and C. Sun, "Attention Bottlenecks for Multimodal Fusion," in *Proc. NeurIPS*, vol. 34, 2021, pp. 14200-14213.